

01 Research profile

Researcher in generative and multimodal intelligence, connecting representation learning and cross-modal reasoning with efficient generative systems.

My work spans speech, sound, music, and broader multimodal learning. First- and co-first-author research has appeared at NeurIPS, ICML, ICLR, ACL, EMNLP, and ACM Multimedia. I work across the research stack – from learning objectives and model architecture to large-scale training, evaluation, and low-latency inference – with **1,000+ citations** and widely adopted open-source releases.

- REPRESENT
Continuous and structured latent spaces
- REASON
Cross-modal planning and reinforcement learning
- GENERATE
Autoregressive, flow-based, and efficient models

02 Selected research agenda

01 AudioCALM

2026

Continuous autoregressive modeling without discrete-token bottlenecks.

Extends next-token prediction to continuous audio latents with a flow-matching head, block-causal AR-Flow, and asymmetric modality experts. **One jointly trained model matches modality-specific state of the art across speech, sound, and music.**

02 STAR-VAE

ICML · 2026

Representation learning for continuous audio generation.

Formalizes the Rate–Distortion–Regularity Trilemma and introduces channel-indexed topology-aware regularization with a CNN–Mamba backbone. **State-of-the-art reconstruction and generation with stronger semantic preservation.**

03 PrismAudio

ICLR · 2026

Reasoning and reinforcement learning for video-to-audio.

Decomposes semantic, temporal, aesthetic, and spatial reasoning into specialized chains of thought and rewards; Fast-GRPO makes multi-objective post-training practical. **State of the art across all four dimensions and strong out-of-domain generalization.**

04 ThinkSound

NEURIPS · 2025

Reasoning before synthesis for interactive audio generation.

A multimodal LLM produces AudioCoT plans for generation, object-centric refinement, and language-guided editing; a unified audio foundation model executes each stage. **Open-source project lead; 1.4K+ GitHub stars and a complete training and inference release.**

05 FlashAudio & AudioLCM

ACL · 2024–25

Moving audio generation from hundreds of steps to one.

Rectified Flow and latent consistency objectives reshape generative trajectories for stable, high-fidelity few-step synthesis. **400× and 333× real-time generation; FlashAudio received ACL Oral and SAC Highlight.**

04 Selected first- and co-first-author work

2026 · UNIVERSAL AUDIO

AudioCALM: Continuous Autoregressive Language Modeling for Universal Audio Generation

Huadai Liu et al. · Continuous autoregressive generation

2026 · ICML

STAR-VAE: Structured Topology-Aware Regularization for Audio Reconstruction and Generation

Huadai Liu et al. · Audio representation learning

2026 · ICLR

PrismAudio: Decomposed Chain-of-Thought and Multi-dimensional Rewards for V2A Generation

Huadai Liu et al. · Multimodal reasoning and RL

2025 · NEURIPS

ThinkSound: Chain-of-Thought Reasoning in Multimodal LLMs for Audio Generation and Editing

Huadai Liu et al. · Audio reasoning and editing

2025 · ICML

OmniAudio: Generating Spatial Audio from 360-Degree Video

Huadai Liu et al. · Spatial audio generation

2025 · ACL

FlashAudio: Rectified Flows for Fast and High-Fidelity Text-to-Audio Generation

Huadai Liu et al. · Oral · SAC Highlight

2025 · ACM MM

MEDIC: Zero-shot Music Editing with Disentangled Inversion Control

Huadai Liu et al. · Oral

2024 · ACM MM

AudioLCM: Efficient and High-Quality Text-to-Audio Generation with Minimal Inference Steps

Huadai Liu et al. · Efficient generation

2024 · ACL FINDINGS

Wav2SQL: Direct Generalizable Speech-To-SQL Parsing

Huadai Liu et al. · Speech reasoning

2023 · EMNLP

ViT-TTS: Visual Text-to-Speech with Scalable Diffusion Transformer

Huadai Liu et al. · Multimodal TTS

Selected collaborative work SpaceVista (ICML 2026), MME-Emotion (ICLR 2026), MaskAudioFlow (Findings of ACL 2026), and one co-authored paper at ECCV 2026. Full record: Google Scholar.

Research & systems impact

5--10× CosyVoice 2 inference

Designed and deployed a Rectified Flow path for substantially faster production inference.

100K+ Training hours

Architected distributed training and multi-domain speech data systems for foundation-model workloads.

22.5K+ CosyVoice

Core contributor to the official open-source ecosystem.

1.4K+ ThinkSound

Project lead; released AudioCoT data, model weights, training and inference code, and demos.

1.2K+ AudioLCM

Project lead; released a widely used baseline for efficient audio generation.

Rounded public repository snapshots, August 2026.

Academic service

ACL Rolling Review

Area Chair · 2026

ECCV

Outstanding Reviewer · 2026

Conference reviewer

ICML · NeurIPS · ICLR · AAAI · ACL · EMNLP · ECCV · ACM Multimedia

Journal reviewer

IEEE/ACM TASLP

Education

Zhejiang University

M.S. Artificial Intelligence · 2021–2024

Top 0.1% of program

Shandong University

B.S. Electronic Information · 2016–2020

Selected honors

ECCV Outstanding Reviewer (2026)

ACL SAC Highlight Award (2025)

National Scholarship, top 1% (2023)

Outstanding Graduate of Zhejiang Province and Zhejiang University (2024)

Technical expertise

RESEARCH

Generative modeling, multimodal LLMs, autoregressive models, flow matching, Rectified Flow, reinforcement learning, and cross-modal reasoning.

ENGINEERING

PyTorch, distributed training, FSDP, DeepSpeed, large-scale data pipelines, streaming inference, and production optimization.